STUDENT CLASSROOM GUIDE

Mixing tank: can a smooth curve be trusted?

What does agreement with an analytic solution check, and what does it leave untested?

50 minutes • Introductory modeling • pairs

Before you begin

Derivatives, units, Python functions and arrays. This is a 50-minute introduction to the existing approximately 150-minute workshop, not its completion.

> The well-mixed tank is an invented teaching system. No measured equipment, mixing validation or plant-safety result is supplied.

One-class sequence

5 min **01. Predict the long-run level**

Sketch a 20 L tank: 2 L/min enters at 10 g/L; equal outflow, reaction 0.05/min, initial concentration 0. Will concentration reach 10 g/L?

10 min **02. State assumptions and balance mass**

Write accumulation as inflow minus outflow minus reaction. Explain why volume is constant and outlet concentration equals tank concentration; check units.

10 min **03. Check two steps, then implement**

Use the initial-value problem below to calculate two Euler steps. Complete student_model and the Euler update in student_euler using your balance.

$$y' = -y, \qquad y(0) = 1, \qquad h = 0.1$$

$$y_{n+1} = y_n + h\, f(t_n, y_n)$$

15 min **04. Run, change one condition, refine**

Run the default, then set reaction to 0.2 per minute. Return to default physical parameters and halve the time step to 0.25 minutes. Record your actual run’ s errors and checks, including failures. If time permits, try zero reaction.

5 min **05. Separate two kinds of evidence**

What would an analytic-solution match tell you? Propose one tracer observation that would challenge the perfect-mixing assumption.

5 min **06. Leave a short evidence record**

Write one result, one check and one limitation. Answer the exit question and export the original workshop’ s Draft JSON or report; do not mark the full workshop complete.

Settings to apply manually

This form updates this classroom page and its share link only. It never loads settings into the activity or changes a saved draft.

Flow Q (L/min)

2

Volume V (L)

20

Reaction k (1/min)

0.05

Inlet concentration (g/L)

10

Initial concentration (g/L)

0

Time step (min)

0.5

Experiment seed

2026

Keep this evidence

- A mass balance with units and three assumptions; two hand-calculated Euler values.
- An actual run or explicitly labelled reference, a refinement comparison and a limitation, exported from the original workshop where available.

> Use the original workshop’ s Draft JSON export/import and report. Drafts stay in this browser; the classroom page never reads or writes them. When storage is unavailable, the workshop labels session-only memory and provides export. Export before changing device.

Exit question

Your numerical curve matches the analytic curve. Does that establish perfect mixing in a real tank? What further evidence is needed?

Printable worksheet

Write on a printed copy or keep notes separately. This page has no answer storage. Use the original activity to save digital work.

My prediction and system boundary:

Balance / units / assumptions:

Two Euler hand values and update rule:

Settings / actual run or reference / error / failed checks:

One result, one check, one limitation and exit response:

Activity source

Sequence adapted from workshop-01 “A tank, a boundary, a balance”, v1.0.0, its existing exercise and reference assets. The 50-minute pacing is an introductory lesson plan; nondimensionalization, failure experiments and extended writing remain in the full workshop.

This guide：https://skckenneth.github.io/teaching/classroom-packs/mixing-tank/
Original activity：https://skckenneth.github.io/teaching/modeling-workshop/mixing-tank/